

Phase 3 — Enhanced Excel Exports

Phase 3 adds a richer, analysis-oriented Excel export layer on top of the validated Daniel-format exporters. It is **purely additive**: `excel_export.py` and `excel_generator_core.py` are never modified, and the `standard` format delegates straight to them so existing deliverables remain identical.

What you get

1. **Color-coded group export** — one stable, colour-blind-safe colour per group (consistent across every sheet), a **Legend** sheet, conditional formatting for over-/under-subscribed groups, alternating row shading, Loyola-branded headers (navy `#003366` / gold `#FFCC00`), frozen panes, auto-filters, and cell comments (professor + room-capacity proxy).
2. **Five analysis sheets** —
 - **Room Utilization** (sessions/students per room, utilisation %, bar chart, conditional formatting `>=80%` green / `50-80%` amber / `<50%` red).
 - **Professor Workload** (sessions, subjects, groups, students, bar chart).
 - **Student Placement** (per-subject status with a global placement banner).
 - **Time Slot Analysis** (Days x Time-blocks heatmap + per-room breakdown).
 - **Quality Metrics** (KPIs from `reports/kpi_report.json` and solver runs from `reports/solver_stats.json`, with traffic-light status).
3. **Advanced filtering & formatting** — auto-filters, dropdown data-validation pickers (Semester / Subject / Professor / Room / Day / Block), workbook-level named ranges, and optional sheet protection that leaves filters usable.
4. **Template system** — branded `.xlsx` templates with `{{TOKEN}}` placeholders (`{{GROUP}}` , `{{PROFESSOR}}` , `{{ROOM}}` , `{{SUBJECT}}` , `{{SEMESTER}}` , `{{DAY}}` , `{{BLOCK}}` , `{{DATE}}` , `{{TITLE}}`), with discovery, validation and rendering.

Formats

Format	Description
<code>standard</code>	Daniel-format workbooks (delegates to <code>excel_export.py</code>). Backward compatible.
<code>summary</code>	Overview + Groups + Legend + Student Placement + Quality Metrics.
<code>enhanced</code>	All sheets, all formatting (default).
<code>detailed</code>	All sheets + protected sheets (filters stay usable).

Using it from the UI

Open the “**Exports avances**” page (`pages/6_Exports_Avancés.py`) in the Streamlit app. Choose a perimeter (both semesters / S1 / S2), a format, a colour scheme (Loyola / Default / Monochrome), tick the sheets and formatting options, preview the data, then **Generer le classeur** and download. The lower section renders a template from user-supplied values.

Using it from code

```
import export_manager as m

# Enhanced workbook for semester 1, Loyola palette.
res = m.export_with_format("enhanced", semester=1, color_scheme="loyola")
print(res["ok"], res["file"])

# Override individual sheets/formatting.
res = m.export_with_format(
    "enhanced", semester=None,
    overrides={"time_slot_analysis": False, "protect_sheets": True},
)

# Render a template.
res = m.render_template(
    "loyola_schedule_template.xlsx",
    {"TITLE": "Distribucion de Practicas", "SEMESTER": "1",
    "SUBJECT": "Fisica", "GROUP": "G1", "PROFESSOR": "Prof",
    "ROOM": "Ciencias Exp. I", "DAY": "Lunes", "BLOCK": "12:30-14:30"},
)
```

Lower-level building (no file I/O) is available via

`excel_export_enhanced.build_enhanced_workbook(...)`, which returns an openpyxl `Workbook`.

Configuration

`config/export_preferences.yaml` controls the default format, default colour scheme, the templates directory, and per-format flag overrides. A missing or invalid file falls back to built-in defaults, so the exporter always works.

Group sizing thresholds

Mirrors `pipeline.py` : min = 7, preferred = 12, max = 15.

- **Optimal:** `7 <= students <= 15` (green `#009E73`)
- **Under-utilized:** `students < 7` (orange `#E69F00`)
- **Over-subscribed:** `students > 15` (vermillion `#D55E00`)

Accessibility

Colours use a colour-blind-safe (Okabe-Ito derived) palette, and status is also encoded by a text label and a fill pattern so meaning survives greyscale and colour-blindness. Per-group colours are deterministic and stable across sheets.

Regenerating the template

```
python templates/build_loyola_template.py
```

This rewrites `templates/loyola_schedule_template.xlsx` deterministically; the generator is kept in the repo so the binary can be reviewed and rebuilt.

Tests

`tests/test_excel_enhanced.py` and `tests/test_export_manager.py` add 52 tests (colour helpers, thresholds, every sheet builder, charts, conditional formatting, comments, data validation, named ranges, format resolution, preferences, the `standard` delegation path, and the template engine). The full suite passes at **174 tests**.